P. E. S. COLLEGE OF ENGINEERING

MANDYA – 571401

(An Autonomous Institution Affiliated to VTU, Belgaum)

MINI PROJECT REPORT

ON

“ UrbanPulseAI – Smart Traffic Intelligence and Decision Support System”

SUBMITTED IN PARTIAL FULFILMENT FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING

IN

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

SUBMITTED BY

AISHWARYA SHASTRI[4PS23CB002] **ASHUTOSH RAJ [4PS23CB005]**

CHARAN B [4PS23CB008] **DIYAGOWDA S S [4PS24CB403]**

UNDER THE GUIDANCE OF

PROF. PUTTASWAMY B S

Assistant Professor

Department of Computer Science and Business Systems

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

2025-2026

P. E. S. COLLEGE OF ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS

CERTIFICATE

This is to certify that Ms. Aishwarya Shastri [4PS23CB002], Mr. Ashutosh Raj [4PS23CB005],Mr. Charan B[4PS23CB008] and Ms. Diya Gowda [4PS24CB403] students of VI semester B.E, **DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS**, P.E.S. College of Engineering Mandya have satisfactorily completed the **MINI PROJECT (P22CBMP607) ON "UrbanPulseAI – Smart Traffic Intelligence and Decision Support System"** during the academic year 2025-26. The Project report has been approved as it satisfies the academic requirements of VI semester B.E Computer Science and Business Systems discipline.

Signature of the Project Guide

Prof. Puttaswamy B S
Assistant Professor
Dept. of CS&BS
PESCE, Mandya

Signature of the HoD

Dr. T M Geethanjali
Head Of The Department
Dept. of CS&BS
PESCE, Mandya

MINI PROJECT EXAMINATION DETAILS			
SL. NO	**NAME OF THE EXAMINER**	**SIGNATURE**	**DATE**
1.			
2.			

DECLARATION

We **Aishwarya Shastri [4PS23CB002], Ashutosh Raj [4PS23CB005], Charan B[4PS23CB008] and Diya Gowda S S [4PS24CB403]** students of 6th semester pursuing **BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND BUSINESS SYSTEMS (CS&BS),** hereby declare that the project report titled **"UrbanPulseAI – Smart Traffic Intelligence and Decision Support System"** is an original work carried out by us during the conduction period. We affirm that this report has not been submitted to any other university or institution for the award of any degree, diploma, or certificate. The information presented in this report is authentic based on our personal experience & learning and submitted in partial fulfilment of the requirements for the award of degree in B.E. Computer Science and Business Systems affiliated to Visvesvaraya Technological University (VTU) Belagavi during the academic year 2025-2026.

DATE:29-06-2026

PLACE:MANDYA

Aishwarya Shastri [4PS23CB002]

Ashutosh Raj [4PS23CB005]

Charan B[4PS23CB008]

Diya Gowda [4PS24CB403]

ACKNOWLEDGEMENT

We express our sincere gratitude to **Dr. N. L. Murali Krishna,** Principal, P.E.S. College of Engineering, Mandya**,** for providing an excellent academic environment and for his constant encouragement and support throughout the course of this project.

We extend our heartfelt thanks to **Dr. T. M. Geethanjali,** Head of the Department of Computer Science and Business Systems**,** for her valuable guidance, encouragement, and continuous support, which greatly contributed to the successful completion of this project.

We would like to express our deepest gratitude to our project guide, **Prof. B. S. Puttaswamy,** Department of Computer Science and Business Systems**,** for his invaluable guidance, constructive suggestions, technical expertise, and constant motivation throughout the development of this project. His encouragement and insightful feedback played a vital role in shaping this work and ensuring its successful completion.

We also express our sincere appreciation to all the faculty members of the Department of Computer Science and Business Systems for their support, encouragement, and valuable academic inputs during the course of this project.

We take this opportunity to thank P.E.S. College of Engineering, Mandya**,** for providing the necessary facilities and resources required to carry out this work successfully.

We gratefully acknowledge everyone who has directly or indirectly contributed to the successful completion of this project titled "UrbanPulseAI – Smart Traffic Intelligence and Decision Support Platform."

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ABSTRACT

Traffic congestion has become one of the most common problems faced by people in modern cities. Increasing numbers of vehicles, rapid urban growth, road accidents, flooding, potholes, and other unexpected incidents often cause delays, inconvenience, and safety concerns for commuters. Traffic authorities also face challenges in identifying problems quickly and responding effectively, especially when information is scattered across multiple sources. As cities continue to grow, there is a need for smarter solutions that can help both citizens and authorities manage transportation more efficiently.

UrbanPulseAI is a smart city platform designed to improve traffic management and urban mobility. The system continuously monitors traffic conditions and helps identify areas where congestion is likely to occur. Citizens can actively participate by reporting incidents such as accidents, road damage, flooding, or road blockages through a simple reporting interface. To improve the reliability of these reports, uploaded images are automatically checked before the complaint is forwarded to the concerned authorities. The platform also helps users find better travel routes, provides traffic updates, and supports authorities with useful information for handling incidents and managing transportation services more effectively.

By bringing together traffic monitoring, citizen participation, incident management, and decision-support capabilities within a single platform, UrbanPulseAI aims to create a safer, smoother, and more efficient transportation environment. The system encourages collaboration between citizens and authorities, enables faster response to road-related problems, and supports better planning and management of urban transportation. Ultimately, it contributes towards building smarter and more sustainable cities while improving the daily travel experience of the public.